The pathophysiologic process of acute myocardial infarction (STEMI) involves a complex relationship between plaque formation due to atherosclerosis and plaque rupture with subsequent thrombus formation leading to an occlusion of one or more of the epicardial coronary arteries and resulting in ischemia of the heart muscle. The patient experienced atherosclerotic plaque rupture leading to thrombus formation and complete coronary artery occlusion, resulting in myocardial ischemia and hypoxia. This caused severe chest pain and ECG changes, decreased cardiac output, and triggered compensatory responses such as tachycardia, hypertension, and diaphoresis. Impaired oxygen delivery led to hypoxaemia and tachypnoea, contributing to the patient's clinical deterioration (Elsaka et al., 2022).

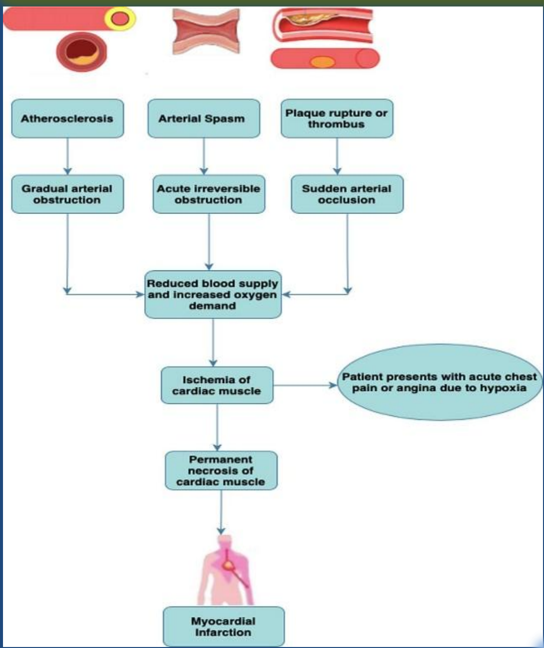

Figure 1: Pathophysiology of STEMI (Source: Botleroo et al., 2021, p. 4)

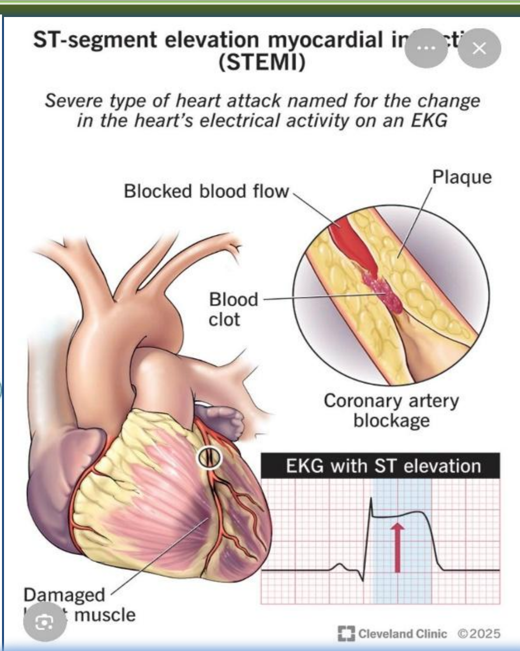

Figure 2: STEMI (Source: Cleveland clinic, 2025)

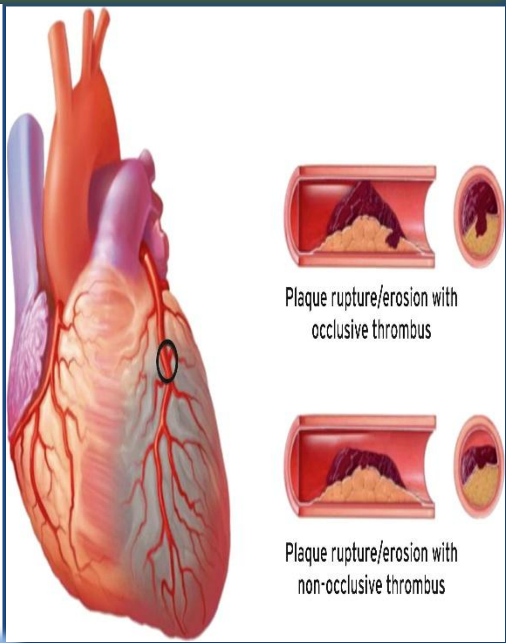

Figure 3: MI (Source: (Thygesen et al., 2018)

Patient X received aspirin after a plaque rupture to prevent platelet aggregation. The aspirin will stop the creation of thromboxane A2 which will help prevent further clotting in the coronary arteries, and it will also help to keep the occlusion from getting worse by improving blood flow through the heart muscle, thus decreasing the amount of heart muscle damaged (Sheu et al., 2026).

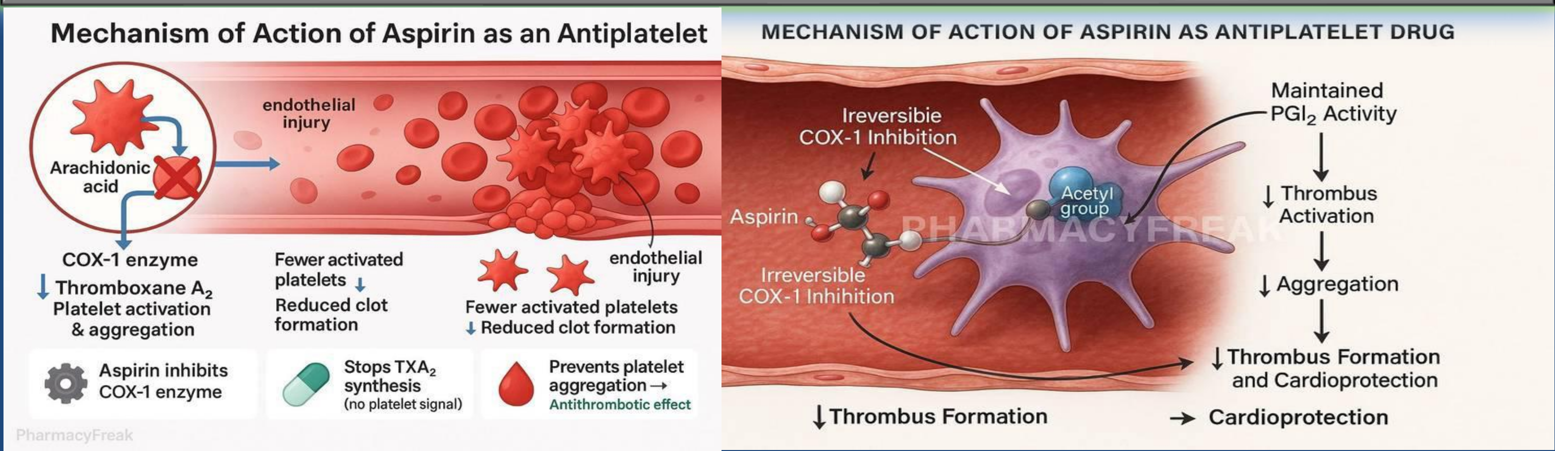

Figure 4&5: Mechanism of Action of Aspirin. (Source: (Rajput, 2025)

The most common adverse effects associated with the use of aspirin are gastrointestinal irritation, an increase in the risk of bleeding, and allergic type reactions. Aspirin should be avoided in patients who have active bleeding, an aspirin allergy, or bleeding disorders. Based on this patient's condition (STEMI), it was appropriate to administer aspirin, but because of her age and potential need for invasive procedures (such as PCI) careful observation and consideration of possible bleeding complications was made (Al Maimani et al., 2025).